



MINISTRY OF EDUCATION
MATRICULATION DIVISION



PHYSICS

LABORATORY MANUAL

**DP014 &
DP024**

5th EDITION



**MATRICULATION DIVISION
MINISTRY OF EDUCATION MALAYSIA**

**PHYSICS
LABORATORY MANUAL
SEMESTER I & II
DP014 & DP024**

**MINISTRY OF EDUCATION MALAYSIA
MATRICULATION PROGRAMME**

FIFTH EDITION

First Printing, 2011 (*First Edition*)
Second Printing, 2015 (*Second Edition*)
Third Printing, 2018 (*Third Edition*)
Fourth Printing, 2020 (*Fourth Edition*)
Fifth Printing, 2022 (*Fifth Edition*)
Copyright © 2022 Matriculation Division
Ministry of Education Malaysia

ALL RIGHTS RESERVED. No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording or any information storage and retrieval system, without the prior written permission from the Director of Matriculation Division, Ministry of Education Malaysia.

Published in Malaysia by

Matriculation Division
Ministry of Education Malaysia,
Level 6 – 7, Block E15,
Government Complex Parcel E,
Federal Government Administrative Centre,
62604 Putrajaya,
MALAYSIA.
Tel : 603-88844083
Fax : 603-88844028
Website : <http://www.moe.gov.my/>

Printed in Malaysia by

Malaysia National Library
Physics Laboratory Manual
Semester I & II
DP014 & DP024
Fifth Edition

e ISBN 978-983-2604-67-9

NATIONAL EDUCATION PHILOSOPHY

Education in Malaysia is an on-going effort towards further developing the potential of individuals in a holistic and integrated manner, so as to produce individuals who are intellectually, spiritually and physically balanced and harmonious based on a firm belief in and devotion to God. Such an effort is designed to produce Malaysian citizens who are knowledgeable and competent, who possess high moral standards and who are responsible and capable of achieving a high level of personal well-being as well as being able to contribute to the betterment of the family, society and the nation at large.

NATIONAL SCIENCE EDUCATION PHILOSOPHY

In consonance with the National Education Philosophy, science education in Malaysia nurtures a science and technology culture by focusing on the development of individuals who are competitive, dynamic, robust and resilient and able to master scientific knowledge and technological competency.

FOREWORD

I am delighted to write the foreword for the Laboratory Manual, which aimed to equip students with knowledge, skills, and the ability to be competitive undergraduates.

This Laboratory Manual is written in such a way to emphasise students' practical skills and their ability to read and understand instructions, making assumptions, apply learnt skills and react effectively in a safe environment. Science process skills such as making accurate observations, taking measurement in correct manner, using appropriate measuring apparatus, inferring, hypothesizing, predicting, interpreting data, and controlling variables are further developed during practical session. The processes are incorporated to help students to enhance their Higher Order Thinking Skills such as analytical, critical and creative thinking skills. These 21st century skills are crucial to prepare students to succeed in Industrial Revolution (I.R.) 4.0.

The manipulative skills such as handling the instruments, setting up the apparatus correctly and drawing the diagrams can be advanced through practical session. The laboratory experiments are designed to encourage students to have enquiry mind. It requires students to participate actively in the science process skills before, during and after the experiment by preparing the pre-report, making observations, analysing the results and in the science process skills before, during, after the experiment by preparing the pre-report, making observations, analysing the results and drawing conclusions.

It is my hope and expectation that this manual will provide an effective learning experience and referenced resource for all students to equip themselves with the skills needed to fulfil the prerequisite requirements in the first-year undergraduate studies.



DR HAJAH ROSNARIZAH BINTI ABDUL HALIM
Director
Matriculation Division

CONTENTS

	Page
1.0 Student Learning Time (SLT)	v
2.0 Learning Outcomes	v
3.0 Guidance for Students	viii
4.0 Significant Figures	xi
5.0 Uncertainty in Measurements	xii

Semester I

Experiment	Title	
1	Physical Measurement	1
2	Plotting and Interpreting Linear Graph	6
3	Free Fall Motion	12
4	Linear Momentum	15
5	Friction	19
6	Thermal Conduction	23

Semester II

Experiment	Title	Page
1	Simple Harmonic Motion (SHM)	26
2	Standing Waves	29
3	Ohm's Law	32
4	Capacitor	35
5	Magnetic Field	38
6	Geometrical Optics	46
References		49
Acknowledgements		50

1.0 Student Learning Time (SLT)

Students will be performing the experiment within the time allocated for each practical work.

Face-to-face	Non face-to-face
2 hours	0

2.0 Learning Outcomes

2.1 Matriculation Science Programme Educational Objectives

Upon a year of graduation from the programme, graduates are:

- i. Knowledgeable and technically competent in science disciplines study in-line with higher educational institution requirement.
- ii. Able to apply information and use data to solve problems in science disciplines.
- iii. Able to communicate competently and collaborate effectively in group work to compete in higher education environment.
- iv. Able to use basic information technologies and engage in life-long learning to continue the acquisition of new knowledge and skills.
- v. Able to demonstrate leadership skills and practice good values and ethics in managing organisations.

2.2 Matriculation Science Programme Learning Outcomes

At the end of the programme, students should be able to:

- i. Acquire knowledge of science and mathematics as a fundamental of higher level education.
(MQF LOC i – Knowledge and understanding)
- ii. Apply logical, analytical and critical thinking in scientific studies and problem solving.
(MQF LOC ii – Cognitive skills)
- iii. Demonstrate manipulative skills in laboratory works.
(MQF LOC iii a – Practical skills)

- iv. Collaborate in group work with skills required for higher education.
(MQF LOC iii b – Interpersonal skills)
- v. Deliver ideas, information, problems and solution in verbal and written communication.
(MQF LOC iii c – Communication skills)
- vi. Use basic digital technology to seek and analyse data for management of information.
(MQF LOC iii d – Digital skills)
- vii. Interpret familiar and uncomplicated numerical data to solve problems.
(MQF LOC iii e – Numeracy skills)
- viii. Demonstrate leadership, autonomy and responsibility in managing organization.
(MQF LOC iii f – Leadership, autonomy and responsibility)
- ix. Initiate self-improvement through independent learning.
(MQF LOC iv – Personal and entrepreneurial skills)
- x. Practice good values attitude, ethics and accountability in STEM and professionalism.
(MQF LOC v – Ethics and professionalism)

2.3 Physics 1 Course Learning Outcome

At the end of the course, student should be able to:

- 1. Describe basic concepts of mechanics and heat.
(C2, PLO 1, MQF LOC i)
- 2. Solve problems related to mechanics and heat.
(C3, PLO 2, MQF LOC ii)
- 3. Apply the appropriate scientific laboratory skills in physics experiments.
(P3, PLO 3, MQF LOC iii a)

2.4 Physics 2 Course Learning Outcome

At the end of the course, student should be able to:

1. Explain basic concepts of waves, electricity, magnetism and optics.
(C2, PLO 1, MQF LOC i)
2. Solve problems of waves, electricity, magnetism and optics.
(C4, PLO 2, MQF LOC ii)
3. Apply the appropriate scientific laboratory skills in physics experiments.
(P3, PLO 3, MQF LOC iii a)

2.5 Physics Practical Learning Outcomes

Physics experiment is to give the students a better understanding of the concepts of physics through experiments. The **aims** of the experiments in this course are to be able to:

1. introduce students to laboratory work and to equip them with the practical skills needed to carry out experiment in the laboratory.
2. determine the best range of readings using appropriate measuring devices.
3. recognise the importance of single and repeated readings in measurement.
4. analyse and interpret experimental data in order to deduce conclusions for the experiments.
5. make conclusions in line with the objective(s) of the experiment which rightfully represents the experimental results.
6. verifying the correct relationships between the physical quantities in the experiments.
7. identify the limitations and accuracy of observations and measurements.
8. familiarise student with standard experimental techniques.
9. choose suitable apparatus and to use it correctly and carefully.

10. gain scientific trainings in observing, measuring, recording and analysing data as well as to determine the uncertainties (errors) of various physical quantities observed in the experiments.
11. handle apparatus, measuring instruments and materials safely and efficiently.
12. present a good scientific report for the experiment.
13. follow instructions and procedures given in the laboratory manual.
14. gain confidence in performing experiments.

3.0 Guidance for Students

3.1 Ethics in the laboratory

- a. Follow the laboratory rules.
- b. Students must be punctual for the practical session. Students are not allowed to leave the laboratory before the practical session ends without permission.
- c. Co-operation between members of the group must be encouraged so that each member can gain experience in handling the apparatus and take part in the discussions about the results of the experiments.
- d. Record the data based on the observations and not based on any assumptions. If the results obtained are different from the theoretical value, state the possible reasons.
- e. Get help from the lecturer or the laboratory assistant should any problems arise during the practical session.

3.2 Preparation for experiment

3.2.1 Planning for the practical

a. Before entering the laboratory

- i) Read and understand the objectives and the theory of the experiment.
- ii) Think and plan the working procedures properly for the whole experiment. Make sure you have appropriate table for the data.
- iii) Prepare a jotter book for the data and observations of the experiments during pre-lab discussion.

b. Inside the laboratory

- i) Check the apparatus provided and note down the important information about the apparatus.
- ii) Arrange the apparatus accordingly.
- iii) Conduct the experiment carefully.
- iv) Record all measurements and observations made during the experiment.

3.3 Report writing

The report must be written properly and clearly in English and explain what has been carried out in the experiment. Each report must contain **name, matriculation number, number of experiment, title, date and practicum group**.

The report must also contain the followings:

- | | | |
|------|-------------|---|
| i) | Objective | <ul style="list-style-type: none">• state clearly |
| ii) | Theory | <ul style="list-style-type: none">• write concisely in your own words• draw and label diagram if necessary |
| iii) | Apparatus | <ul style="list-style-type: none">• name, range, and sensitivity, e.g
Voltmeter: 0.0 – 10.0 V
Sensitivity: ± 0.1 V |
| iv) | Procedure | <ul style="list-style-type: none">• write in passive sentences about all the steps taken during the experiment |
| v) | Observation | <ul style="list-style-type: none">• data tabulation with units and uncertainties• data processing (plotting graph, calculation to obtain the results of the experiments and its uncertainties).• Calculation of uncertainties using LSM method can refer attachment A |
| vi) | Discussion | <ul style="list-style-type: none">• give comments about the experimental results by comparing it with the standard value• state the source of mistake(s) or error(s) if any as well as any precaution(s) taken to overcome them• answer all the questions given |
| vii) | Conclusion | <ul style="list-style-type: none">• state <u>briefly</u> the results with reference to the objectives of the experiment |

Reminder: NO PLAGIARISM IS ALLOWED.

4.0 Significant Figures

The significant figures of a number are those digits carry meaning contributing to its precision. Therefore, the most basic way to indicate the precision of a quantity is to write it with the correct number of significant figures.

The significant figures are all the digits that are known accurately plus the one estimated digit. For example, we say the distance between two towns is 200 km, that does not mean we know the distance to be exactly 200 km. Rather, the distance is 200 km *to the nearest kilometres*. If instead we say that the distance is 200.0 km that would indicate that we know the distance to the nearest *tenth* of a kilometre.

More significant figures mean greater precision.

Rules for identifying significant figures:

1. Nonzero digits are always significant.
2. Final or ending zeros written to the right of the decimal point are significant.
3. Zeros written on either side of the decimal point for the purpose of spacing the decimal point are not significant.
4. Zeros written between significant figures are significant.

Example:

Value	Number of significant figures	Remarks
0.5	1	Implies value between 0.45 and 0.55
0.500	3	Implies value between 0.4995 and 0.5005
0.050	2	Implies value between 0.0495 and 0.0505
5.0	2	Implies value between 4.95 and 5.05
1.52	3	Implies value between 1.515 and 1.525
1.52×10^4	3	Implies value between 15150 and 15250
150	2 or 3 (ambiguous)	The zero may or may not be significant. If the zero is significant, the value implied is between 149.5 and 150.5. If the zero is not significant, the value implied is between 145 and 155.

5.0 Uncertainty in Measurements

No matter how careful or how accurate are the instruments, the results of any measurements made at best are only close enough to their true values (actual values). Obviously, this is because the instruments have certain smallest scale by which measurement can be made. Chances are, the true values lie within the smallest scale. Hence, we have uncertainties in our measurements.

The uncertainty of a measurement depends on its type and how it is done. For a quantity x with uncertainty Δx , the measurement should be recorded as $x \pm \Delta x$ with appropriate unit.

The relative uncertainty of the measurement is defined as $\frac{\Delta x}{x}$.

and therefore its percentage of uncertainty, is given by $\frac{\Delta x}{x} \times 100\%$.

5.1 Single Reading

- (a) If the reading is taken from a single point or at the end of the scale we use:

$$\Delta x = \frac{1}{2} \times (\text{smallest division of the scale})$$

- (b) If the readings are taken from two points on the scale:

$$\Delta x = 2 \times \left[\frac{1}{2} \times (\text{smallest division from the scale}) \right]$$

- (c) If the apparatus has a vernier scale:

$$\Delta x = 1 \times (\text{smallest unit of the vernier scale})$$

5.2 Repeated Readings

For a set of n repeated measurements, the best value is the average value, that is

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

where: n is the number of measurements taken
 x_i is the i^{th} measurement value

The uncertainty is given by

$$\Delta x = \frac{\sum_{i=1}^n |\bar{x} - x_i|}{n}$$

The result should be written in the form of

$$x = \bar{x} \pm \Delta x$$

5.3 Combination of uncertainties

We adopt maximum uncertainty.

- (a) Addition or subtraction

$$x = a + b - c \Rightarrow \Delta x = \Delta a + \Delta b + \Delta c$$

- (b) Multiplication with constant k

$$x = ka \Rightarrow \Delta x = k\Delta a$$

- (c) Multiplication or division

$$x = \frac{ab}{c} \Rightarrow \frac{\Delta x}{x} = \left(\frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c} \right)$$

- (d) Powers

$$x = a^n \Rightarrow \frac{\Delta x}{x} = n \left(\frac{\Delta a}{a} \right)$$

5.4 Uncertainty gradient and y-intercept using Least Square Method (LSM)

5.4.1 Formula uncertainty for gradient and y-intercept

Straight line graphs are very useful in data analysis for many physics experiments.

From straight line equation, that is, $y = mx + c$ we can easily determine the gradient m of the graph and its intercept c on the vertical axis.

When plotting a straight line graph, the line does not necessary passes through all the points. Therefore, it is important to determine the uncertainties Δm and Δc for the gradient of the graph and the y-interception respectively.

Consider the data obtained is as follows:

x	x_1	x_2	$x_3, \dots, x_n$
y	y_1	y_2	$y_3, \dots, y_n$

- (a) Find the centroid $(\bar{x}, \bar{y})$, where

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad \text{and} \quad \bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

- (b) Draw the best straight line passing through the centroid and balance.
- (c) Determine the gradient of the line by drawing a triangle using dotted lines. The gradient is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

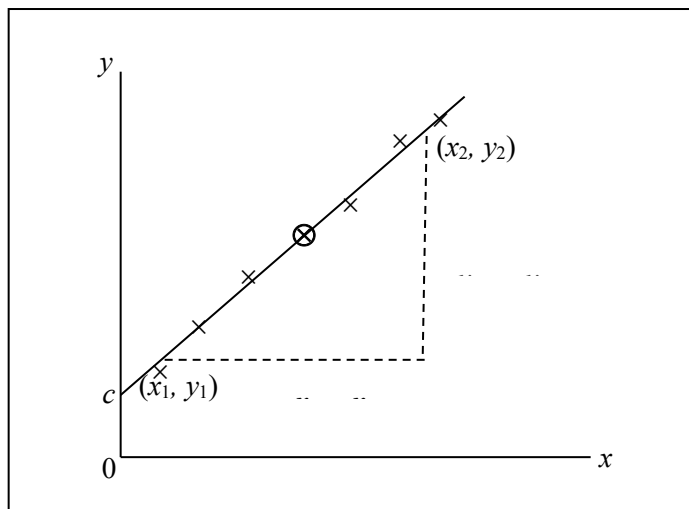


Figure A

- (d) The uncertainty of the slope, Δm can be calculated using the following equation

$$\Delta m = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{(n-2) \sum_{i=1}^n (x_i - \bar{x})^2}}$$

where n is the number of readings and $\bar{x}$ is the average value of x given by

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

and the estimated value of y , $\hat{y}_i$ is given by,

$$\hat{y}_i = \hat{m}x_i + \hat{c}$$

- (e) The uncertainty of the y-intercept, Δc can be calculated using the following equation

$$\Delta c = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)}$$

5.4.2 Procedure to draw a straight line graph and to determine its gradient with its uncertainty

- Choose appropriate scales to use at least 80% of the sectional paper. Draw, label, mark the two axes, and give the units. Avoid using scales of 3, 7, 9, and the likes or any multiple of them. Doing so will cause difficulty in plotting the points later on.
- Plot all points clearly with $\times$. At this stage you can see the pattern of the distribution of the graph points. If there is a point which is clearly too far-off from the rest, it is necessary to repeat the measurement or omit it.
- Calculate the centroid and plot it on the graph.

Example:

Suppose a set of data is obtained as below. Graph of T^2 against ℓ is to be plotted.

ℓ (± 0.1 cm)	10.0	20.0	30.0	40.0	50.0	60.0
T^2 (± 0.01 s ²)	0.33	0.80	1.31	1.61	2.01	2.26

From the data:

$$\bar{\ell} = \frac{10.0 + 20.0 + 30.0 + 40.0 + 50.0 + 60.0}{6} = 35.0 \text{ cm}$$

$$\overline{T^2} = \frac{0.33 + 0.80 + 1.31 + 1.61 + 2.01 + 2.26}{6} = 1.39 \text{ s}^2$$

Therefore, the centroid is (35.0 cm, 1.39 s²).

- (d) Draw a best straight line through the centroid and balance. Points above the line are roughly in equal number and positions to those below the line.
- (e) Determine the gradient of the line. Draw a fairly large right-angle triangle with part of the line as the hypotenuse.

From the graph in **Figure B**, the gradient of the line is as follows:

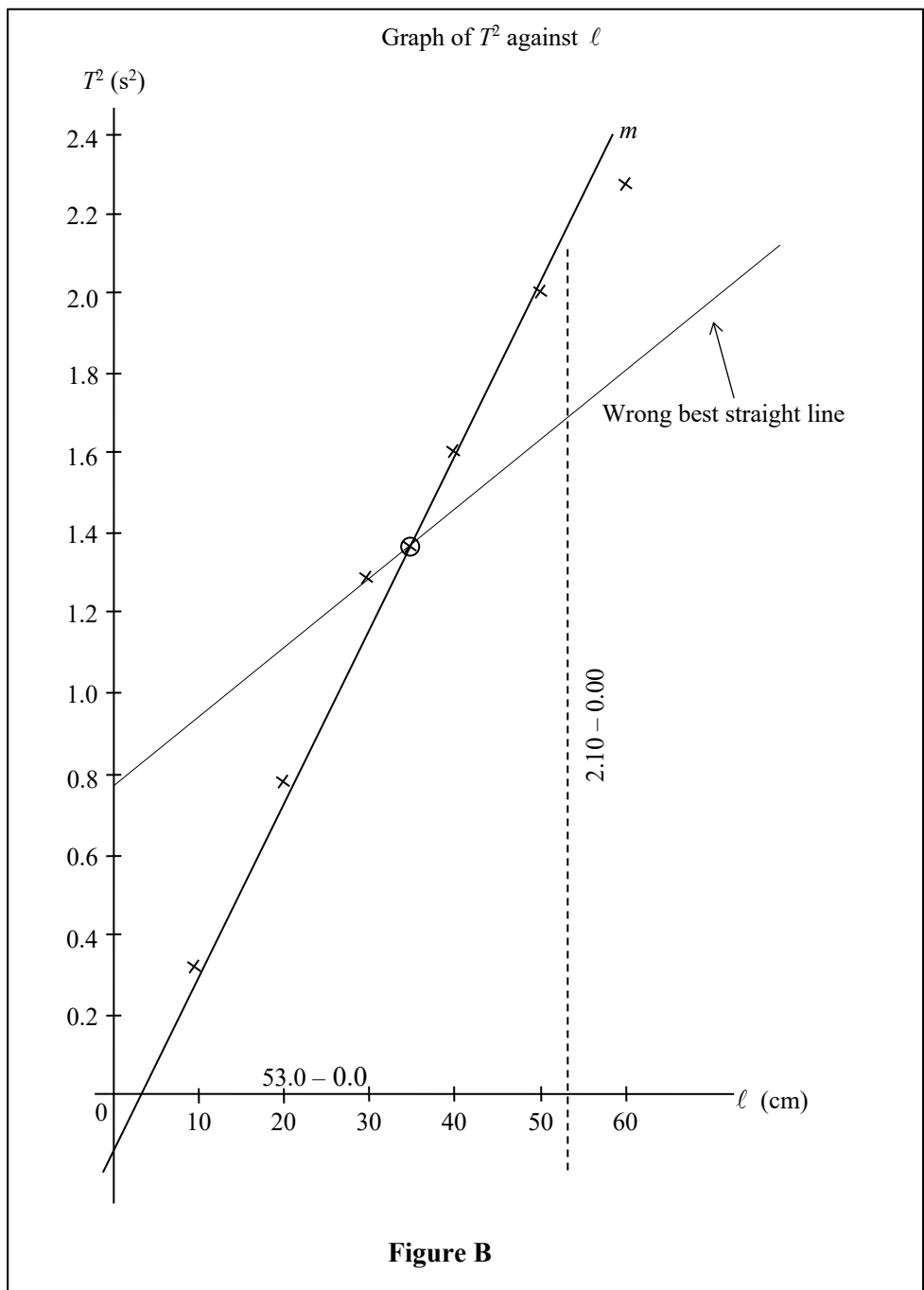
For the best line:

$$\begin{aligned} m &= \frac{(2.10 - 0.00) \text{ s}^2}{(53.0 - 0.0) \text{ cm}} \\ &= 0.040 \text{ s}^2 \text{ cm}^{-1} \end{aligned}$$

The gradient of the graph and its uncertainty should be written as follows:

$$m = (0.040 \pm \text{___}) \text{ s}^2 \text{ cm}^{-1}$$

Take extra precaution so that the number of significant figures for the gradient and its uncertainty are in consistency.



(f) Calculation of uncertainties

Rewrite the data in the form of

ℓ	$\ell - \bar{\ell}$	$(\ell - \bar{\ell})^2$	T^2	$\widehat{T^2}$	$T^2 - \widehat{T^2}$	$(T^2 - \widehat{T^2})^2$
10.0	-25.0	625.0	0.33	0.4	-0.070	0.0049
20.0	-15.0	225.0	0.80	0.8	0.000	0.0000
30.0	-5.0	25.0	1.31	1.2	0.110	0.0121
40.0	5.0	25.0	1.61	1.6	0.010	0.0001
50.0	15.0	225.0	2.01	2.0	0.010	0.0001
60.0	25.0	625.0	2.26	2.4	-0.140	0.0196
$\Sigma=210.0$		$\Sigma=1750.0$				$\Sigma=0.0368$

Where, $\bar{\ell}$ is the average of ℓ ,

$$\bar{\ell} = \frac{210}{6} = 35.0 \text{ cm}$$

Where, $\widehat{T^2}$ is the expected value of T^2

$$\widehat{T^2} = 0.04\ell$$

Calculate the uncertainty of slope, Δm

$$\begin{aligned} \Delta m &= \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{(n-2) \sum_{i=1}^n (x_i - \bar{x})^2}} \\ &= \sqrt{\frac{0.0368}{(6-2)(1750)}} \\ &= \pm 0.002 \end{aligned}$$

Then, calculate the uncertainty of y-intercept, Δc

$$\begin{aligned} \Delta c &= \sqrt{\left(\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2} \right) \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)} \\ \Delta c &= \sqrt{\left(\frac{0.0368}{6-2} \right) \left(\frac{1}{6} + \frac{35^2}{1750} \right)} \\ \Delta c &= \pm 0.09 \end{aligned}$$

The data given in section 5.4.2(e) was obtained from an experiment to verify the relation between T^2 and ℓ . Theoretically, the quantities obey the following relation,

$$T^2 = \left(\frac{k}{p} \right) \ell$$

where k is a natural number equals 39.48 and p is a physical constant. Calculate p and its uncertainty.

Solution:

From the equation, we know that

$$\begin{aligned} \frac{k}{p} &= \text{gradient } m \\ p &= \frac{k}{m} \\ &= \frac{39.48}{0.040} \\ &= 987 \text{ cm s}^{-2} \end{aligned}$$

Since k is a natural number which has no uncertainties, that is $\Delta k = 0$.

$$\begin{aligned} \Delta p &= \left(\frac{\Delta k}{k} + \frac{\Delta m}{m} \right) p \\ &= \left(0 + \frac{0.002}{0.040} \right) 987 \\ &= 49.35 \end{aligned}$$

so we write,

$$p = (987 \pm 49.35) \text{ cm s}^{-2} \quad \text{or} \quad p = (1000 \pm 50) \text{ cm s}^{-2}$$

5.5 Percentage of difference:

When comparing an experimental result to a value determined by theory or to an accepted known value, the difference between the experimental value and the theoretical value can be determined by:

$$\text{Percentage of difference} = \left| \frac{X_{\text{Theory}} - X_{\text{Experiment}}}{X_{\text{Theory}}} \right| \times 100\%$$

SEMESTER I

EXPERIMENT 1: PHYSICAL MEASUREMENT

Objective:

To measure and determine the uncertainty of physical quantities.

Theory:

Measuring some physical quantities is part and parcel of any physics experiment. It is important to realise that not all measured values are exactly the same as the actual values. This could be due to the errors that we made during the measurement, or perhaps the apparatus that we used may not be accurate or sensitive enough. Therefore, as a rule, the uncertainty of a measurement must be taken and it has to be recorded together with the measured value.

The uncertainty of a measurement depends on the type of measurement and how it is done. For a quantity x with the uncertainty Δx , its measurement is recorded as below:

$$x \pm \Delta x$$

The relative uncertainty of the measurement is defined as:

$$\frac{\Delta x}{x}$$

and therefore, its percentage of uncertainty is given by $\frac{\Delta x}{x} \times 100\%$

1. Single Reading

- 1.1 If the reading is taken from a single point or at the end of the scale,

$$\Delta x = \frac{1}{2} \times (\text{the smallest division from the scale})$$

- 1.2 If the readings are taken from two points on the scale,

$$\Delta x = 2 \times \left[\frac{1}{2} \times (\text{the smallest division from the scale}) \right]$$

- 1.3 If the apparatus uses a vernier scale,

$$\Delta x = 1 \times (\text{the smallest unit from the vernier scale})$$

2. Repeated Readings

For a set of n repeated measurements of x , the best value is the average value

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad 1.1$$

where n = the number of measurements taken

x_i = the i^{th} measurement

The uncertainty is given by

$$\Delta x = \sum_{i=1}^n \frac{|\bar{x} - x_i|}{n} \quad 1.2$$

The result should be written as

$$x = \bar{x} \pm \Delta x \quad 1.3$$

3. Combination of uncertainties

We adopt maximum uncertainty.

3.1 Addition or subtraction

$$x = a \pm b \pm c \quad \Rightarrow \quad \Delta x = \Delta a + \Delta b + \Delta c$$

3.2 Multiplication with constant k

$$x = ka \quad \Rightarrow \quad \Delta x = k\Delta a$$

3.3 Multiplication or division

$$x = \frac{ab}{c} \quad \Rightarrow \quad \Delta x = \left(\frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c} \right) x$$

3.4 Powers

$$x = a^n \quad \Rightarrow \quad \Delta x = n \left(\frac{\Delta a}{a} \right) x$$

Apparatus:

A micrometer screw gauge
 A vernier calipers
 A metre rule
 A ball bearing
 A coin
 A metal rod
 A glass rod
 An electronic balance

Procedure:

1. Choose the appropriate instrument for measurement of single reading
 - i. The length of a metal rod.
 - ii. The length and width of a laboratory manual.
 - iii. The mass of a ball bearing.
2. Determine the percentage of uncertainty for each set of readings.
3. Choose the appropriate instrument for measurement repeated reading
 - i. The diameter of a ball bearing.
 - ii. The diameter a coin.
 - iii. The external diameter of a glass rod.
4. For procedure 3, perform the measurement and record the data in a suitable table for at least 5 readings. (Refer to **Table 1.1** as an example)

Table 1.1

No.	The diameter of a ball bearing, d ($\pm \dots \dots \text{cm}$)	$ \bar{d} - d_i \text{ cm}$
1		
2		
3		
4		
5		
Average	$d = \frac{\sum_{i=1}^n d_i}{n} =$	$\Delta d = \frac{\sum_{i=1}^n \bar{d} - d_i }{n} =$

5. Determine the percentage of uncertainty for each set of readings.

6. Calculate the following derived quantities and its uncertainties.
 - i. Perimeter of the laboratory manual.
 - ii. Circumference of the coin
 - iii. Surface area of the ball bearing.
 - iv. Volume of the ball bearing.
 - v. Density of the ball bearing.

EXPERIMENT 2: PLOTTING AND INTERPRETING LINEAR GRAPH

Objective:

To develop skill in plotting and interpreting linear graph.

Theory:

Graphs are often used to represent the dependence or relationship of two or more experimental quantities. In experimental work, the independent variable is usually represented by the x -axis while the dependent variable is represented by the y -axis. The shape of the graphs can take many forms, but our focus will only be on linear graphs. Linear graphs obey the following equation

$$y = mx + c \quad 2.1$$

where m is the gradient of the graph and c is the intercept on the y -axis.

1. Procedure to draw a graph

1.1 Axes scale, label and title

- a) Choose the scale on each axis such that all data points should be shown and should fill not less than half the size of the plotted area. The scales should be in multiples of 10, 5, 2 or 1 and DO NOT USE odd multiples. In some cases, it may be necessary to adjust the graph with an axis that does not start at zero.
- b) Label the scales of each axis with simple numbers. In cases where the numbers are very large or very small, use scientific notation.
- c) Axes title should represent the relevant quantities with appropriate units.

1.2 Plotting the data points and the straight line

- a) Determine the centroid which is defined as follows:

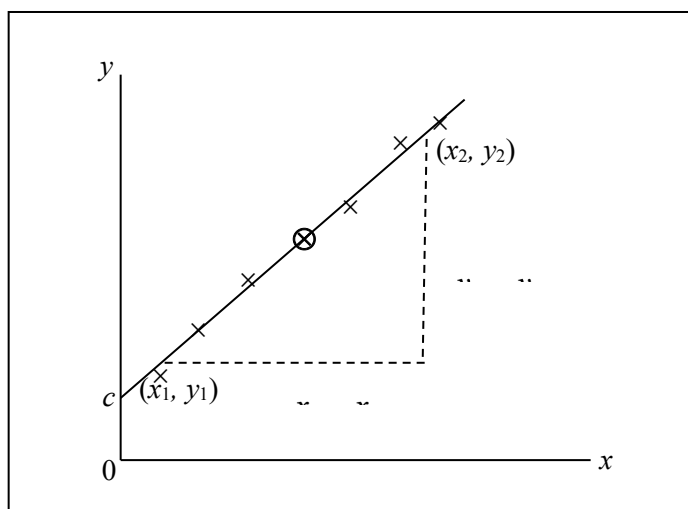
$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad \bar{y} = \frac{\sum_{i=1}^n y_i}{n} \quad 2.2$$

- b) The data points should be plotted using the symbol $\times$. Use the symbol $\otimes$ to plot the centroid.
- c) Draw a best fit solid straight line. The line must pass through the centroid and as many data points as possible as shown in Figure 2.1. It may not necessarily pass through every data point and the origin.

1.3 Determination of gradient

- a) The gradient of the graph is determined by choosing two points (x_1, y_1) and (x_2, y_2) which lie on the line (NOT THE DATA POINTS) and could be read accurately. Draw a right-angle triangle using these two points.
- b) The value of the gradient is calculated as:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad 2.3$$

**Figure 2.1**

Note: The size of the triangle is such that $(x_2 - x_1)$ is greater than 50% of the x -axis data range (refer Figure 2.1).

2. Transformation to linear graph

The relationship between physical quantities may not be linear. However, the relationship often can be transformed into a linear form and hence a linear graph may be plotted. For example, the relationship between the period, T of oscillation of a simple pendulum of length ℓ is given as

$$T = 2\pi \sqrt{\frac{\ell}{g}} \quad 2.4$$

which can be expressed in linear form as

$$T^2 = \frac{4\pi^2}{g} \ell \quad 2.5$$

Thus, a plot of T^2 against ℓ is a straight line passing through the origin with gradient given by

$$m = \frac{4\pi^2}{g} \quad 2.6$$

Exercise

1. **Table 2.1** shows data taken in a free fall experiment. Measurements were made of the distance of fall (y) at each precisely measured times by using digital stopwatch.

Table 2.1

Distance, y (± 0.1 cm)	Time, t (± 0.001 s)			$\bar{t}$ (s)	t^2 (s ²)
	t_1	t_2	t_3		
10.0	0.143	0.139	0.140		
20.0	0.202	0.199	0.200		
30.0	0.247	0.245	0.249		
40.0	0.286	0.286	0.280		
50.0	0.319	0.320	0.317		
60.0	0.350	0.346	0.352		
70.0	0.378	0.380	0.376		

- a) Complete the table. Use proper number of significant figures in the table entries.
- b) The equation of motion for an object in free fall starting from rest is $y = \frac{1}{2}gt^2$, where g is the acceleration due to gravity. This is a parabolic equation, which has the general form $y = ax^2$.
 - i. Convert the curve to a straight line by plotting a graph of y against t^2 .
 - ii. Determine the gradient of the line and calculate the experimental value of g .

2. A physical pendulum which is made of a rod has its axis of oscillation at a distance, h from its centre of mass as shown in **Figure 2.2**. The period of oscillation, T is tabulated in **Table 2.2**.

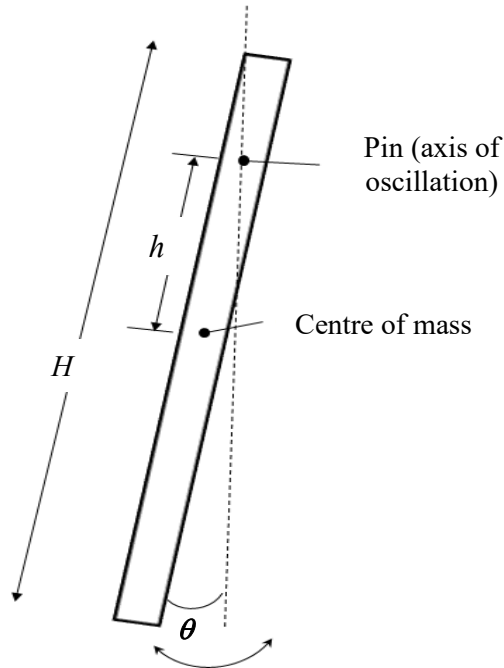


Figure 2.2

Table 2.2

h (± 0.1 cm)	20.0	40.0	60.0	80.0	100.0	120.0	140.0	160.0
T (± 0.01 s)	2.65	2.54	2.39	2.20	2.00	1.83	1.55	1.26
T^2 (s^2)								

The period of oscillation is given as

$$T = 2\pi \sqrt{\frac{H - h}{g}} \quad 2.7$$

where H is a constant and g is the acceleration due to gravity.

- a) Complete Table 2.2
- b) Express equation 2.7 in linear form as equation 2.1.
- c) Plot a linear graph to determine H and g .

EXPERIMENT 3: FREE FALL MOTION

Objective:

To determine the acceleration due to gravity, g using free fall motion

Theory:

When a body of mass, m falls freely from a certain height, h above the ground, it experiences a linear motion. The body will obey the equation of motion,

$$s = ut + \frac{1}{2}at^2 \quad 3.1$$

By substituting,

$s = -h$ = downward displacement of the body from the falling point to the ground

$u = 0$ = the initial velocity of the body

$a = -g$ = the acceleration due to gravity

t = time taken for the body to reach the ground

we obtain the displacement of the body, h as

$$h = \frac{1}{2}gt^2 \quad 3.2$$

Apparatus:

A retort stand with a clamp

A timer

A metre rule

A free fall adaptor

A horizontal table

A steel ball

Procedure:

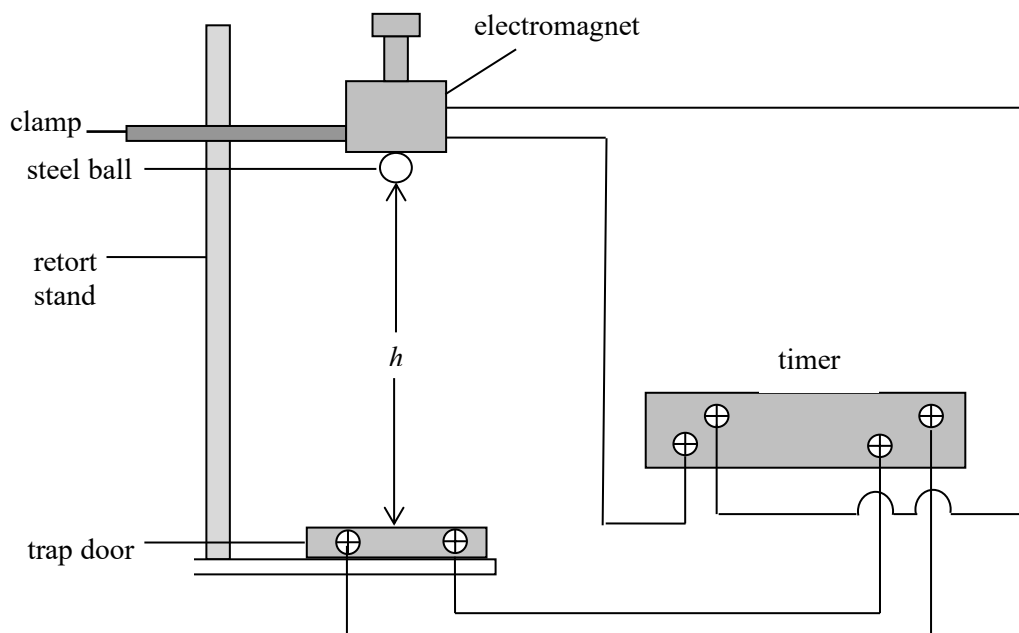


Figure 3.1

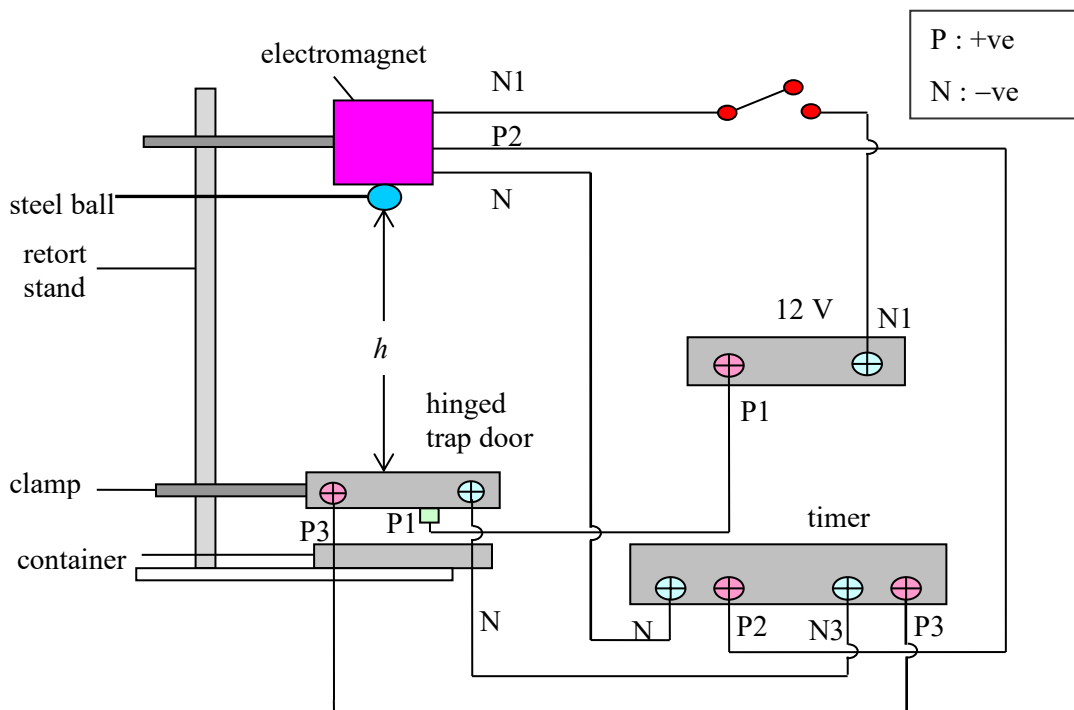


Figure 3.2

1. Set up the apparatus as in **Figure 3.1 or Figure 3.2** (Set up for free fall apparatus with separate power supply to electromagnet).
2. Switch on the circuit and attach the steel ball onto the upper contact.
3. Adjust the height of the electromagnet above the point of impact.
4. Begin with a small value of h .
5. Switch off the circuit and let the ball fall. Record the value of h and t .
6. Take six sets of reading at different values of h and t .
7. Plot a graph of h against t^2 .
8. Calculate the gradient of the graph and determine the value of acceleration due to gravity, g .
9. Determine the uncertainty of acceleration due to gravity, Δg .
10. Compare the acceleration due to gravity, g obtained from this experiment with the standard value. Write the comments.

EXPERIMENT 4: LINEAR MOMENTUM

Objective:

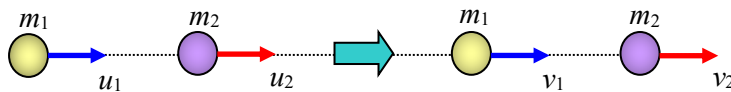
To verify the principle of conservation of linear momentum.

Theory:

Linear momentum, p is a vector quantity defined as the product of mass and velocity. For a body of mass, m and velocity, v ; its momentum is given as $\vec{p} = m\vec{v}$. In the absence of external force, the total momentum of the system is conserved. This is known as the Principle of Conservation of Linear Momentum.

Applying the principle of conservation of linear momentum in collision, the total momentum of the colliding bodies remains the same before and after the collision. Let m_1 and m_2 be the masses of two colliding bodies (i.e: body 1 and body 2) as in **Figure 4.1** and $\vec{u}_1$, $\vec{u}_2$, $\vec{v}_1$, and $\vec{v}_2$ are the velocities before and after the collision of body 1 and body 2 respectively and written as

$$m_1\vec{u}_1 + m_2\vec{u}_2 = m_1\vec{v}_1 + m_2\vec{v}_2 \quad 4.1$$



Collision in 1D

Figure 4.1

In this experiment, we shall study the collision between two ball bearings on a curved track. The first ball bearing is released from the top of the track so that it gains momentum $m_1\vec{u}_1$ before it hits the second ball bearing which is stationary at the horizontal end of the track.

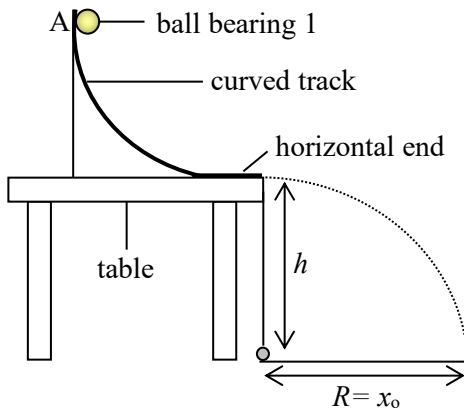


Figure 4.2a

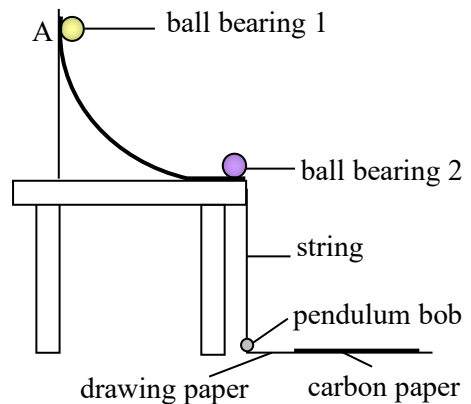


Figure 4.2b

Consider the collision in x-axis only. Assume that the velocity of the ball bearing is directly proportional to its horizontal displacement, R .

$$m_1x_0 = m_1x_1 + m_2x_2 \quad 4.2$$

Therefore, if equation 4.2 practically satisfied, the principle of conservation of linear momentum is thus verified.

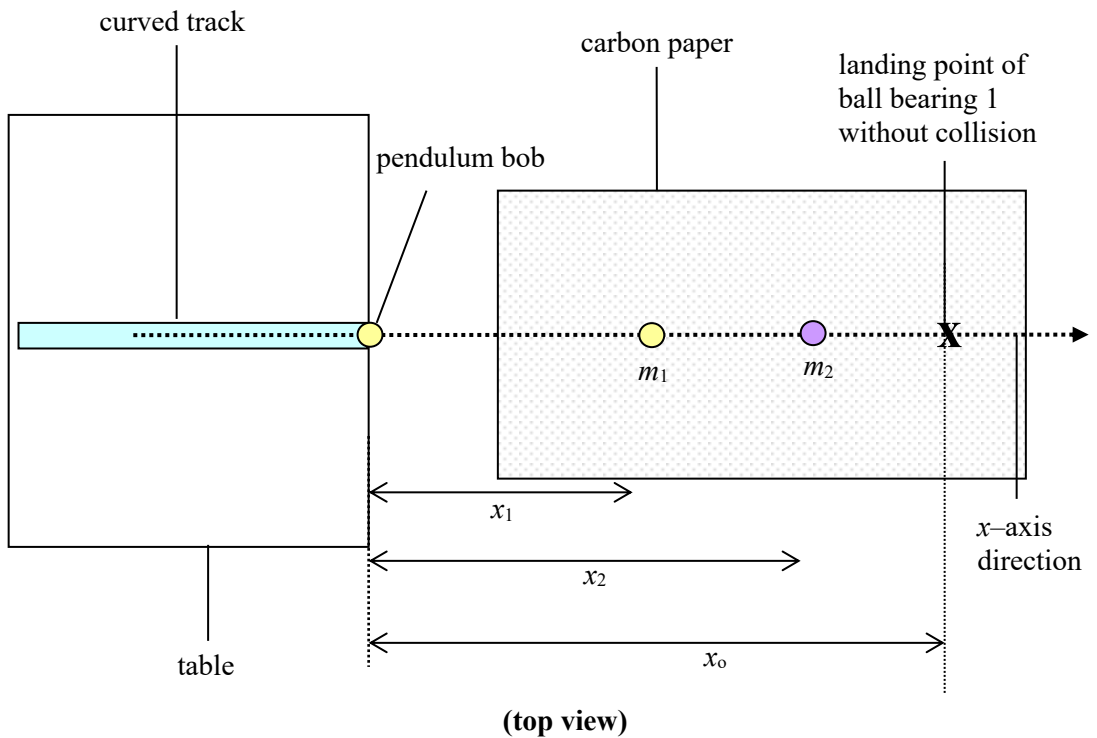


Figure 4.3

Apparatus:

A level table

A curved track.

(Important: The lower end of the track must be horizontal)

2 ball bearings

A piece of string

A pendulum bob

A metre rule

A piece of drawing paper (A3)

A piece of carbon paper (A4)

A retort stand

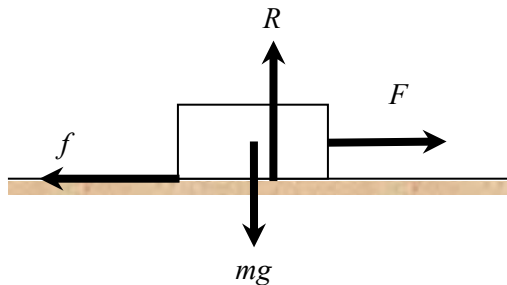
Procedure:

1. Weigh the mass of the two ball bearings, m_1 and m_2 .
2. Set up the apparatus as in **Figure 4.2a**.
3. Mark point A on the curved track as in **Figure 4.2a**.
4. Release ball bearing 1 from point A.
5. Observe the landing point of ball bearing 1 on the floor in order to place the carbon paper on top of a drawing paper so as to mark accurately the point where the ball bearing will land on the floor.
6. Release ball bearing 1 from point A and measure the distance from the bottom end of the pendulum bob to the landing point of ball bearing 1, x_0 . Repeat this step to obtain at least five readings and calculate the average value of x_0 .
7. Place ball bearing 2 on the end of the horizontal track, the path of ball bearing 1 so that one dimension (1D) collision occurs.
Note: *Make sure the landing point of both ball bearings are along the x -axis.*
8. Release ball bearing 1 from point A (Refer to **Figure 4.2b**). These two ball bearings will subsequently collide and fall freely in a projectile motion.
9. Measure the position of landing points for each ball bearing. x_1 for ball bearing 1 and x_2 for ball bearing 2.
(Refer to **Figure 4.3**)
10. Repeat step (7) to (9) for at least another four sets.
11. Does your result satisfy equations 4.2? Give comments.

EXPERIMENT 5: FRICTION

Objective:

To determine the coefficients of static friction and kinetic friction.

Theory:**Static friction (at the onset of motion):****Figure 5.1**

By referring to **Figure 5.1**, if force F is increased, frictional force f also increases accordingly and the object still remain at rest. However, for a certain value of F , the object starts to move. At this stage, the frictional force is known as the limiting static frictional force f_s which is the maximum value of f . Hence,

$$f_s = \mu_s R \quad 5.1$$

$$f_s = \mu_s mg \quad 5.2$$

where, μ_s = coefficient of static friction
 f_s = static frictional force
 R = normal reaction force

Kinetic friction (in motion):

Now, if the object is in motion, the frictional force is known as kinetic friction f_k . The kinetic frictional force is less than the static frictional force. That explains why it is difficult to move an object which is initially at rest, but once it is in motion, less force is needed to maintain the motion.

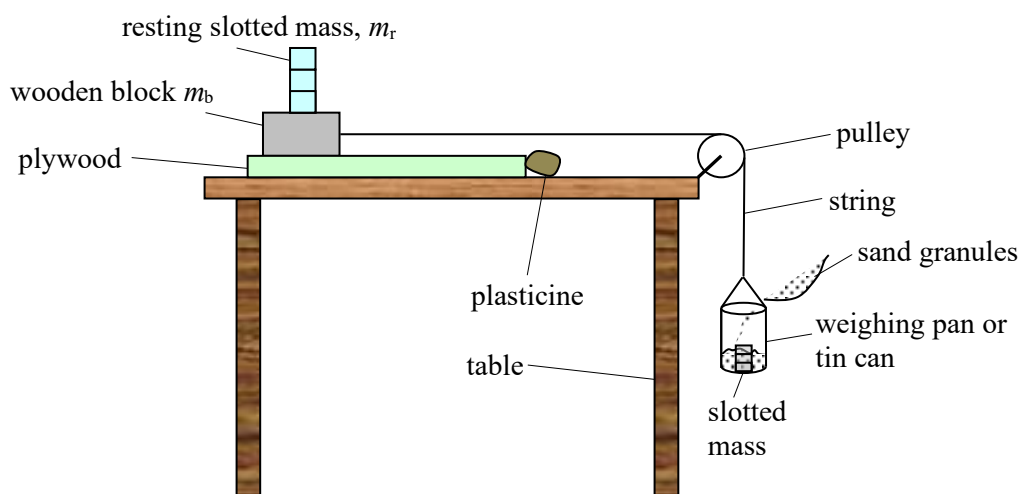
$$f_k = \mu_k R \quad 5.3$$

where μ_k = coefficient of kinetic friction
 f_k = kinetic frictional force
 R = normal reaction force

Since $f_k < f_s$, therefore $\mu_s > \mu_k$.

Apparatus:

A piece of plywood
 A wooden block
 Two sets of slotted mass (2 g, 5 g, 10 g, 20 g)
 Sand granules or small metal pellets (each about 2 g)
 Weighing pan or a tin can
 A set of pulley with clamp
 A piece of string
 Plasticine
 An electronic balance

Procedure:**Figure 5.2**

1. Weigh the mass of the wooden block, m_b .
2. Set up the apparatus as in **Figure 5.2**. Make sure that the string from the block is tied up horizontally to the pulley. Mark the initial position of the wooden block.
3. Add the slotted mass into the weighing pan or tin can gradually until the wooden block begins to slip. If the wooden block still remains static, begin adding the sand granules or the metal pallets gradually until the block starts to move.

Note: *Add the mass gently to avoid impulsive force.*

4. Weigh and record the total mass, m_h required to move the wooden block. Repeat this step three times to get the average value of m_h .
5. Add different masses of m_r onto the wooden block and repeat step (3) to (5).
6. Repeat step (5) for at least five different values of m_r .

7. Plot a graph of f_s against R where $f_s = m_h g$ and $R = (m_r + m_b)g$.
8. Calculate the gradient of the graph and determine μ_s , the coefficient of static friction from the graph.
9. Repeat step (3) but exert a little push (using tic-tac pen) to the wooden block every time each mass is added. Weigh and record the total mass, m_h when the block moves slowly and steadily along the plywood.
10. Add different masses of m_r onto the wooden block and repeat step (9).
11. Repeat step (10) for at least five different values of m_r .
12. Plot a graph of f_k against R where $f_k = m_h g$ and $R = (m_r + m_b)g$.
13. Calculate the gradient of the graph and determine μ_k , the coefficient of kinetic friction from the graph.
14. Determine the uncertainty of both μ_s and μ_k .
15. Compare the value of μ_s and μ_k . Does this confirm to the theory?

EXPERIMENT 6: THERMAL CONDUCTION

Objective:

To determine the thermal conductivity of glass.

Theory:

Thermal conductivity k could be expressed in terms of the rate of flow of heat,

$$\frac{dQ}{dt} = -kA \frac{dT}{dx} \quad 6.1$$

where

$$\frac{dQ}{dt} = \text{rate of heat flow}$$

A = tangential surface area for heat flow / Cross-sectional area

$$\frac{dT}{dx} = \text{temperature gradient}$$

Relationship between temperature T and time t for this experiment is given by

$$\log T_0 - \log T = \frac{kt}{Brx} \quad 6.2$$

or,

$$\log T = -\frac{kt}{Brx} + \log T_0 \quad 6.3$$

where T = temperature (in Kelvin)

$T_0 = 293 \text{ K}$

t = time

$B = 4.84 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$ (specific heat capacity of water)

r = average radius of the boiling tube

x = thickness of the wall of the boiling tube

Apparatus:

A boiling tube
A digital thermometer
A mercury thermometer
A 1000 cm³ beaker
Two stirrers
A cork
A stopwatch
A vernier calipers
A retort stand and clamp
Ice cubes
Hot water

Procedure:

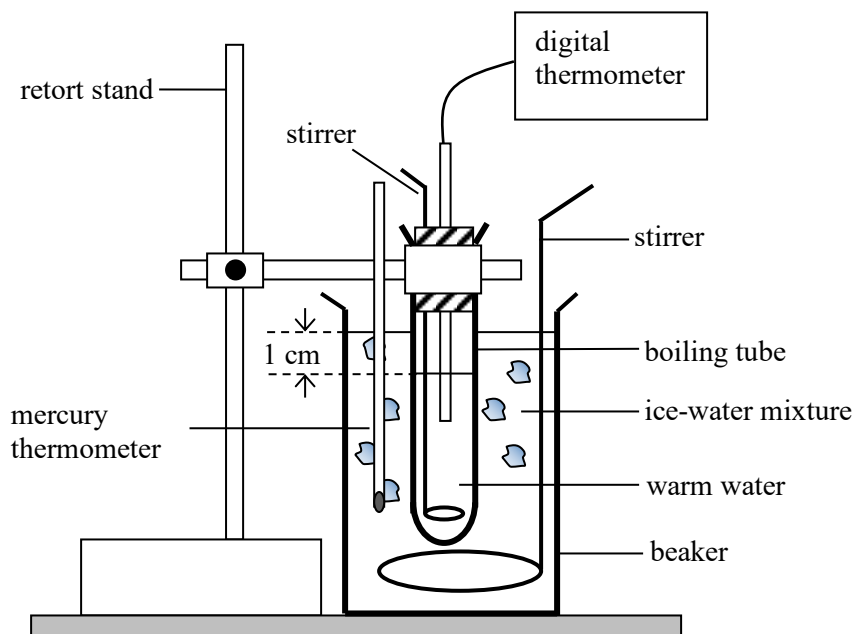


Figure 6.1

1. Measure the internal and external diameters of a boiling tube and calculate the average radius r and the thickness x of the wall of the boiling tube.
2. Fill up a beaker with water and ice.
3. By referring to **Figure 6.1**, clamp the boiling tube on a retort stand and lower the boiling tube into the beaker until the whole of the boiling tube almost submerge in the ice and water mixture. The temperature of the ice-water mixture inside the beaker should be 0°C .
4. Pour hot water into the boiling tube until the water level inside the tube reaches about 1 cm below the ice-water level in the beaker.
5. Insert the stirrer and thermometer through the cork.
6. Record at regular time t and the corresponding temperature T starting from 30°C until it drops to 3°C . The ice-water mixture in the beaker and the warm water in the boiling tube should be constantly stirred throughout the experiment.
7. Tabulate t , T and $\log T$.
8. Plot a graph of $\log T$ against t .
9. Calculate the gradient of the graph and determine the thermal conductivity of glass, k .
10. Determine the uncertainty of k .
11. Compare the result with the standard value $k = 0.8 \text{ W m}^{-1} \text{ K}^{-1}$ for glass. Give comments.

SEMESTER II

EXPERIMENT 1 : SIMPLE HARMONIC MOTION (SHM)

Objective:

To determine the acceleration, g due to gravity using a simple pendulum.

Theory:

An oscillation of a simple pendulum is a simple harmonic motion if:

- (i) the mass of the spherical bob is a point mass
- (ii) the mass of the string is negligible
- (iii) amplitude of the oscillation is small ($< 10^\circ$)

According to the theory of SHM, the period of oscillation of a simple pendulum, T is given as:

$$T = 2\pi \sqrt{\frac{l}{g}} \quad 1.1$$

where l is the length of pendulum
 g is the acceleration due to gravity

Rearrange equation 1.1, we obtained

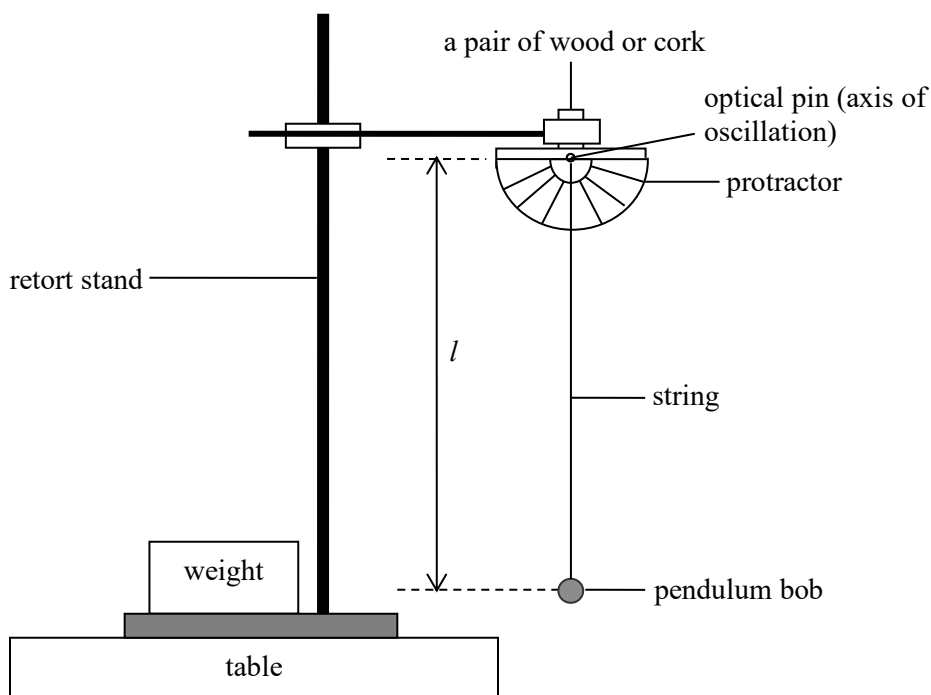
$$T^2 = \frac{4\pi^2 l}{g} \quad 1.2$$

Evidently, a graph of T^2 against l is a straight line of gradient equals $\frac{4\pi^2}{g}$.

Hence, from the gradient of the graph, the value of g can be calculated.

Apparatus:

A piece of string (≈ 105 cm)
 A small pendulum bob
 A pair of small flat pieces of wood or cork
 A retort stand with a clamp
 A stopwatch
 A metre rule
 A protractor with a hole at the centre of the semicircle
 An optical pin
 A pair of scissors or a cutter
 A stabilizing weight or a G-clamp

Procedure:**Figure 1.1**

1. Set up a simple pendulum as in **Figure 1.1**.
2. Measure the length, l of the pendulum at 40 cm.
3. Release the pendulum at less than 10° from the vertical in one plane and measure the time, t for 10 complete oscillations.

Note: *Start the stopwatch after several complete oscillations.*
4. Calculate the average time, t .
5. Calculate the period of oscillation, T of the pendulum.
6. Repeat step (3) to (5) for length, l of 50 cm, 60 cm, 70 cm, 80 cm and 90 cm.
7. Tabulate the data.
8. Plot a graph of T^2 against l .
9. Determine the acceleration due to gravity, g from the gradient of the graph.
10. Compare the acceleration due to gravity, g obtained from step (9) with the standard value. Write a comment.

EXPERIMENT 2: STANDING WAVES

Objectives:

- (i) To investigate standing waves formed in a stretched string.
- (ii) To determine the mass per unit length, μ of the string.

Theory:

When a stretched string is vibrated at a frequency, f the standing waves formed have both ends as nodes. The frequency in the string obeys the following relation

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

Hence, the tension of the string,

$$T = 4\mu f^2 l^2 \quad 2.1$$

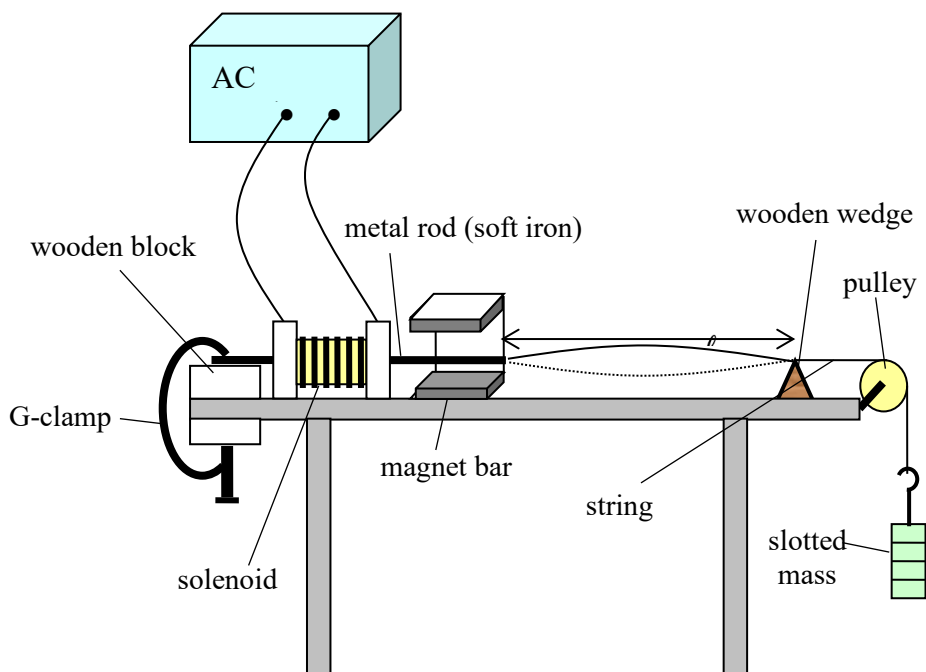
where f is the frequency
 l is the length between two nodes
 T is the tension in the string
 μ is the mass per unit length

Evidently, a graph of T against l^2 is a straight line of gradient equals $4\mu f^2$.

Hence, the mass per unit length, μ can be calculated.

Apparatus:

A G-clamp
 A solenoid (about 100 turns) or ticker timer
 An AC supply (2 – 4 V)
 A metal rod (soft iron)
 Two bar magnets
 A magnet holder
 A piece of string approximately 2 m long
 A pulley with clamp
 A wooden wedge
 Set of slotted mass 10 g
 A metre rule
 Connecting wires

Procedure:**Figure 2.1**

1. Set up the apparatus as in **Figure 2.1**.
2. Connect the terminals of the solenoid to the AC power supply (2 V, 50 Hz).

Caution: Do not exceed 4 V to avoid damage to the solenoid.
3. Place the metal rod between the two bar magnets.
4. Tie one end of the string to the rod and the other to the hook of the slotted mass. Make sure that the length of the string from the end of the rod to the pulley is **not less than 1.5 m**.
5. Clamp the metal rod properly. Switch on the power supply. Adjust the position of the metal rod to get maximum vibration.
6. Place the wooden wedge below the string **as close as possible to the pulley**.

7. Adjust the position of the wooden wedge until a clear single loop standing wave (fundamental mode) is observed.
8. Record the distance, l between the wooden wedge and total mass, m (mass of the hook and the slotted mass). Calculate weight, W where $W = mg$.

Note: *Weight, $W = \text{Tension, } T$*

9. Add 10 g slotted mass to the hook and repeat step (7) and (8) for at least six different readings.
10. Tabulate the data.
11. Plot a graph of T against l^2 .
12. Determine the mass per unit length, μ from the gradient of the graph if the frequency of the vibration is 50 Hz.
13. Weigh the mass of the string and measure the total length of the string. Calculate the mass per unit length, μ of the string.
14. Compare the mass per unit length, μ in step (12) with the result obtained in step (13). Write a comment.

EXPERIMENT 3: OHM'S LAW

Objectives:

- (i) To verify Ohm's law.
- (ii) To determine the effective resistance of the resistors arranged in series and parallel.

Theory:

At constant temperature the potential difference, V across a conductor is directly proportional to the current, I that flows through it. The constant of proportionality is known as resistance of the conductor denoted by R . Mathematically,

$$V \propto I$$

$$V = IR \quad 3.1$$

For resistors in series, the effective resistance is

$$R = R_1 + R_2 + R_3 + \dots \quad 3.2$$

For resistors in parallel, the effective resistance is

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \quad 3.3$$

Apparatus:

A DC power supply (4 – 6 V)

Three resistors of the same resistance (27 -100 Ω)

A DC milliammeter

A DC ammeter (1 A)

A DC voltmeter

A resistance box

A switch

Connecting wires

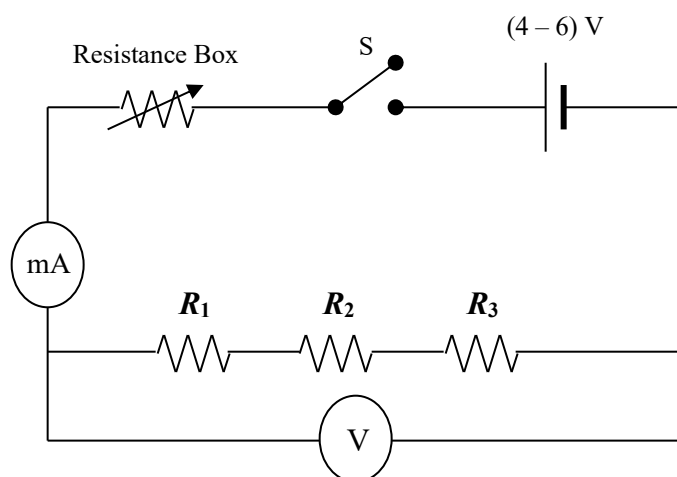
Procedure:

Figure 3.1

1. Determine the resistance, R of each resistor from their colour bands.
2. Set up the circuit as in **Figure 3.1**. Connect the three resistors in series.

Note: *Ask your lecturer to check the circuit before switching ON the power.*

3. Change the resistance value in the resistance box to get a minimum reading in the milliammeter. Record the reading of the voltmeter, V and the milliammeter, I .
4. Change the resistance value in the resistance box to obtain at least five different values of V and I .

5. Tabulate the data.
6. Plot a graph of V against I .
7. Determine the effective resistance, R_{eff} of the three resistors connected in series from the gradient of graph.
8. Set up the circuit as in **Figure 3.2**. Connect the three resistors in parallel.

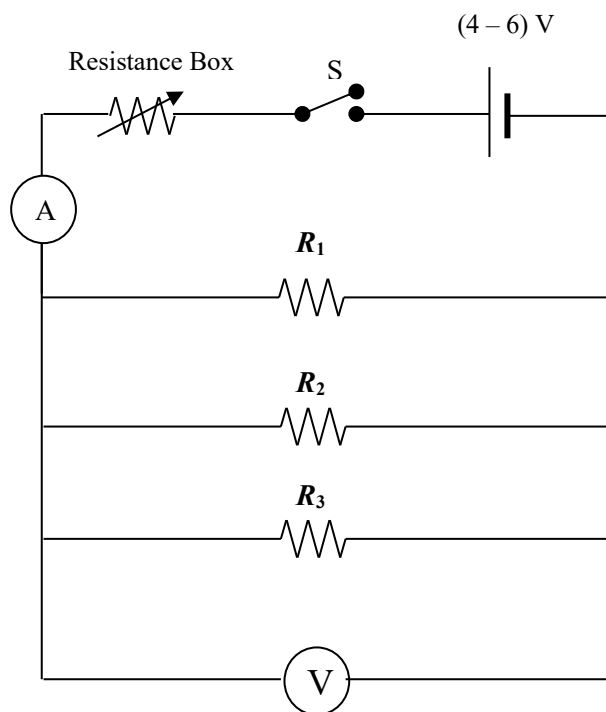


Figure 3.2

9. Repeat steps (3) to (6).
10. Determine the effective resistance, R_{eff} of the three resistors connected in parallel from the gradient of the graph.
11. Compare all the results with their respective calculated values.
12. Verify Ohm's law from the plotted graphs. Write a comment.

EXPERIMENT 4: CAPACITOR

Objective:

- (i) To determine the time constant, τ of an RC circuit.
- (ii) To determine the capacitance, C of a capacitor using an RC circuit.

Theory:

Time constant is defined as the time taken of a discharge current in capacitor decreases to 37% of its maximum current. The time constant can be calculated by using

$$\tau = RC \quad 4.1$$

where τ is the time constant
 R is the resistance of a resistor
 C is the capacitance of a capacitor

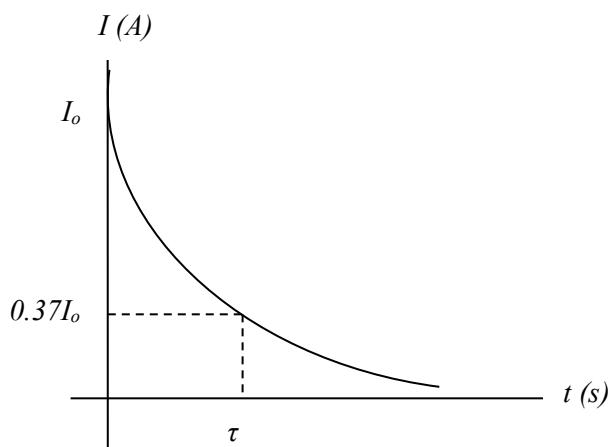


Figure 4.1

During discharging, the magnitude of the current, I varies with time, t as shown in **Figure 4.1**.

From Figure 4.1, the magnitude of the discharge current is

$$I = I_0 e^{-\frac{t}{\tau}} \quad 4.2$$

where I_0 is the maximum current in the circuit
 I is the current in the circuit at time t

Apparatus:

A DC power supply (4 – 6 V)
 A switch
 A DC microammeter
 A digital stopwatch
 A 100 k Ω resistor
 Connecting wires
 A capacitor labelled C (470 – 1000 μF)

Procedure:

Note: Before starting or repeating this experiment, make sure that the capacitors are fully discharged. This can be attained by short circuiting the capacitors.

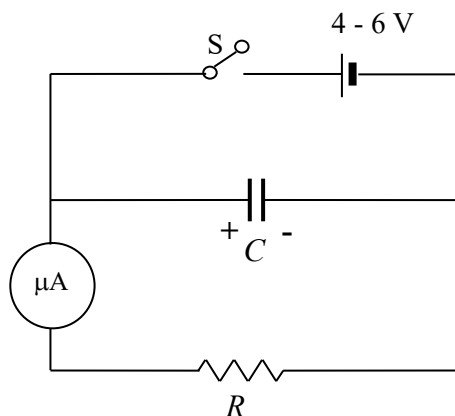


Figure 4.2

1. Set up the circuit as shown in **Figure 4.2**.
2. Close switch S to fully charge the capacitor C. Record the reading of the microammeter for maximum current, I_o .
3. Open switch S and start the stopwatch simultaneously.
4. Use the 'lap' function on the stopwatch when the current reaches at least six different values. Record the time for each value of current.
5. Repeat steps (1) to (4) and calculate the average time, t .
6. Tabulate the values for both I and t .
7. Plot graph of I against t .
8. Determine the time constant, τ from the graph.
9. Calculate the capacitance of the capacitor C by using equation 4.1.
10. Compare the value from step (9) with the actual value. Write a comment.

EXPERIMENT 5 (a): MAGNETIC FIELD

Objective:

To determine the value of the horizontal component of the earth magnetic field $\vec{B}_E$.

Theory:

The magnetic field strength $\vec{B}$ is a vector quantity so the addition of two magnetic fields obeys the parallelogram law. For example, if $\vec{B}_E$ is the horizontal component of earth magnetic field and $\vec{B}_C$ is the magnetic field of a coil which is perpendicular to $\vec{B}_E$ then the resultant of the two fields $\vec{B}$ is as shown in **Figure 5.1**. A compass needle is situated at the place where the two fields meet will be aligned to the direction of the resultant field $\vec{B}$.

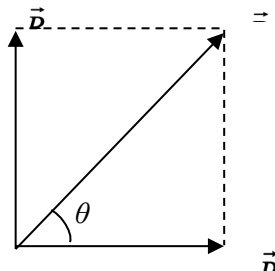


Figure 5.1

From Biot-Savart's Law, the magnetic field strength of the coil at the centre as shown in **Figure 5.2** is given by

$$B_c = \frac{\mu_o NI}{D} \quad 5.1$$

where $\mu_o = 4\pi \times 10^{-7} \text{ H m}^{-1}$ (permeability of free space)

I is the current in ampere

N is the number of turns in the coil

D is the diameter of the coil

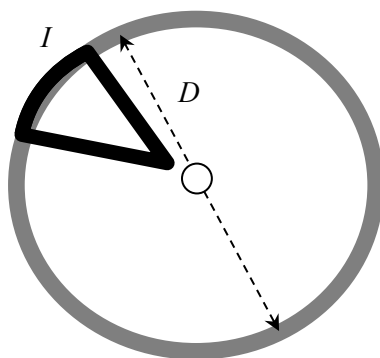


Figure 5.2

From Figure 5.1,

$$\tan \theta = \frac{B_c}{B_E}$$

$$\tan \theta = \frac{\mu_0 N}{D(B_E)} I$$
5.2

The gradient of graph $\tan \theta$ against I

$$m = \frac{\mu_0 N}{D(B_E)}$$
5.3

Therefore,

$$B_E = \frac{\mu_0 N}{Dm}$$
5.4

Apparatus:

Earth magnetic field measurement kit (EM-01)

Connecting wires of about 50 cm long with crocodile clips

A DC ammeter (0 – 1 A)

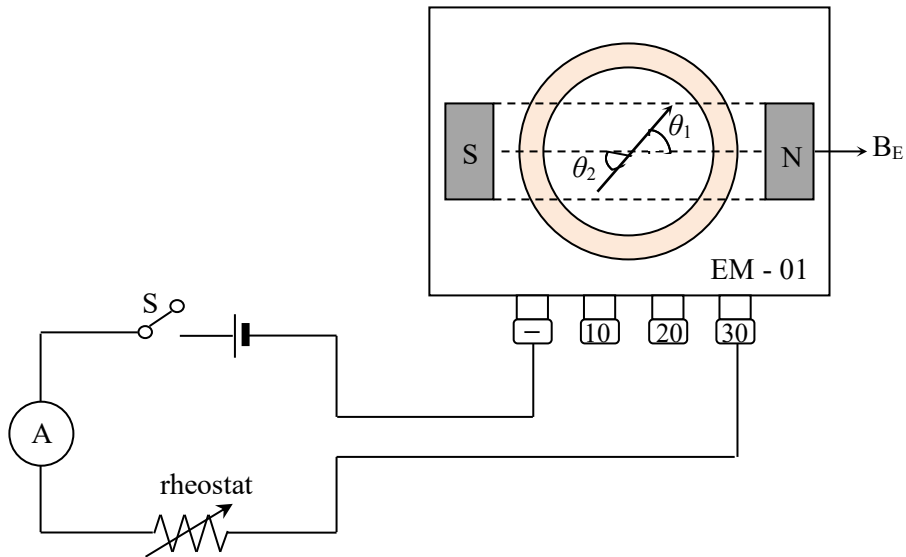
A rheostat

A DC power supply (2 - 4 V)

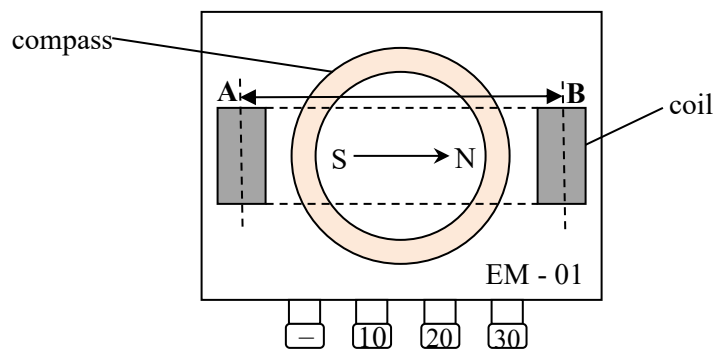
Procedure:

1. Set up apparatus as shown in **Figure 5.3**.

Note: Make sure the EM-01 kit is located far away from other electrical devices to avoid magnetic disturbance.

**Figure 5.3**

2. Align the compass needle of EM-01 until it is pointed North as in **Figure 5.4**.

**Figure 5.4**

- Set the rheostat to its maximum value and switch on the circuit. Reduce the resistance of rheostat to increase the current I and hence the corresponding value of θ_1 for at least six sets of readings. Record the readings of the ammeter I and the angle of deflection θ_1 in **Table 5.1**.

Note: The deflection angle should not be more than 80° .

- Repeat step (3) by changing the polarity of the power supply. Record the angle θ_2 , pointed by the compass needle in **Table 5.1**.
- Measure the diameter, D of the coil.

Note: Make sure the diameter is measured from A to B as in **Figure 5.4**

Diameter of coil $D = (\dots\dots\dots \pm \dots\dots\dots)$ cm

Number of turns $N = 30$

Table 5.1

No.	current, I ($\pm\dots\dots\dots$)	θ_1 ($\pm\dots\dots\dots$)	θ_2 ($\pm\dots\dots\dots$)	average θ_A ($\dots\dots$)	$\tan \theta_A$
1					
2					
3					
4					
5					
6					

- Plot a graph of $\tan \theta$ against I .
- Determine B_E from the gradient of the graph.
- Compare the result with the standard value given by the lecturer. Write a comment.

EXPERIMENT 5 (b): MAGNETIC FIELD

Objective:

To determine the value of the horizontal component of the earth magnetic field, $\vec{B}_E$.

Theory:

The magnetic field strength $\vec{B}$ is a vector quantity so the addition of two magnetic fields obeys the parallelogram law. For example, if $\vec{B}_E$ is the horizontal component of earth magnetic field and $\vec{B}_s$ is the magnetic field of a solenoid which is perpendicular to $\vec{B}_E$ then the resultant of the two fields $\vec{B}$ is as shown in **Figure 5.1**. A compass needle is situated at the place where the two fields meet will be aligned to the direction of the resultant field $\vec{B}$.

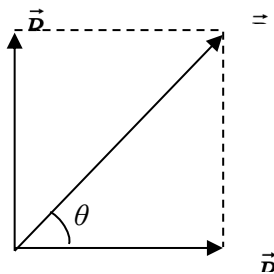


Figure 5.1

The magnetic field strength at the end of an N -turn solenoid of length l and carries current I as shown in **Figure 5.2** is given by

$$B_s = \frac{1}{2} \left(\frac{\mu_0 N I}{l} \right) \quad 5.1$$

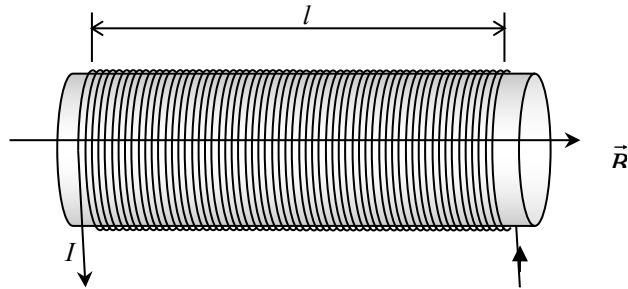


Figure 5.2

From **Figure 5.1**,

$$\begin{aligned}\tan \theta &= \frac{B_s}{B_E} \\ \tan \theta &= \frac{\frac{1}{2} \mu_o \left(\frac{N}{l} \right) I}{B_E}\end{aligned}\quad 5.2$$

The gradient of graph $\tan \theta$ against I is

$$m = \frac{\frac{1}{2} \mu_o \left(\frac{N}{l} \right)}{B_E} \quad 5.3$$

Therefore,

$$B_E = \frac{\frac{1}{2} \mu_o \left(\frac{N}{l} \right)}{lm} \quad 5.4$$

where $\mu_o = 4\pi \times 10^{-7} \text{ H m}^{-1}$ (permeability of free space).

I is the current in ampere

N is the number of turns in the coil

D is the diameter of the coil

Apparatus:

A 50 turns or 100 turns solenoid

A DC power supply (2 – 4 V)

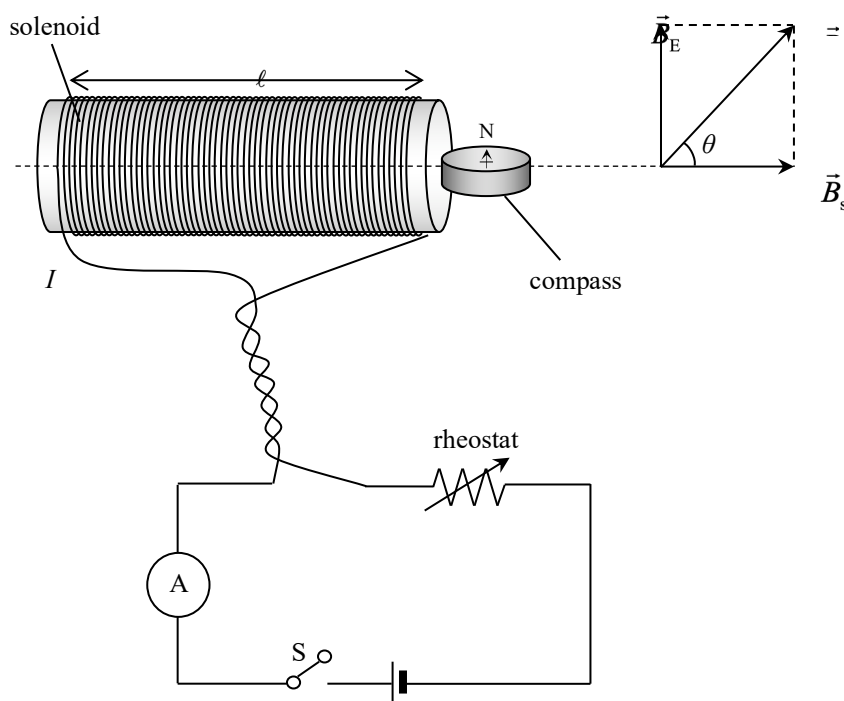
A DC ammeter (0 – 1 A)

A switch

Connecting wires of about 50 cm long with crocodile clips

A rheostat

A compass

Procedure:**Figure 5.3**

1. Place a compass at one end of the solenoid. Let the compass stay still in N–S direction where the magnet pointer is perpendicular to the axis of the solenoid. The north direction of the compass must be pointed to the north.

Note: Choose a position to place your compass away from any iron structure to avoid any influence on the alignment of the compass needle.

2. Connect the solenoid in series with the rheostat, the ammeter, the power supply and the switch. The ammeter must be at least 50 cm away from the compass. A complete set up is as in **Figure 5.3**.
3. Set the rheostat to its maximum value and switch on the circuit. Reduce the resistance of rheostat to increase the current I and hence the corresponding value of θ_1 for at least six sets of readings. Record the readings of the ammeter I and the angle of deflection θ_1 in **Table 5.1**.

Note: The deflection angle should not be more than 80° .

4. Repeat step (3) by changing the polarity of the power supply. Record the angle θ_2 , pointed by the compass needle in **Table 5.1**.

Table 5.1

No.	Current, I ($\pm \dots\dots\dots$)	θ_1 ($\pm \dots\dots\dots$)	θ_2 ($\pm \dots\dots\dots$)	Average θ_A ($\dots\dots$)	$\tan \theta_A$
1					
2					
3					
4					
5					
6					

5. Remove the solenoid from the clamp and measure the length, l of the solenoid.
6. Plot a graph of $\tan \theta$ against I .
7. Determine B_E from the gradient of the graph.
8. Compare the result with the standard value given by the lecturer. Write a comment.

EXPERIMENT 6: GEOMETRICAL OPTICS

Objective:

To determine the focal length, f of a convex lens.

Theory:

From the lens equation,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad 6.1$$

where,

f is the focal length

u is the object distance

v is the image distance

Multiply equation 6.1 with v , gives

$$\frac{v}{f} = \frac{v}{u} + 1$$

$$m = -\frac{v}{f} + 1 \quad 6.2$$

where $m = \frac{\text{height of image}}{\text{height of object}} = \frac{h_i}{h_o} = -\frac{v}{u}$ is the linear magnification.

For this experiment the image formed is always real, and then a negative sign for magnification indicates that the image is inverted.

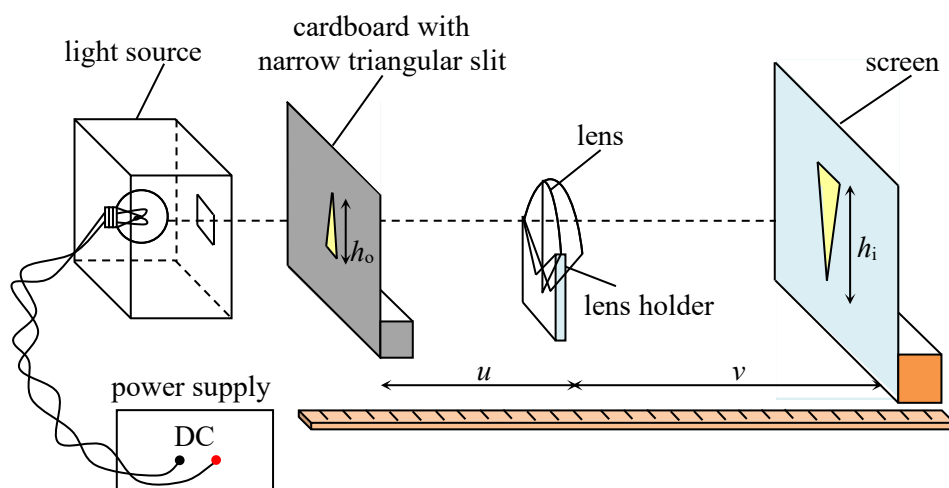
Hence, the graph m against v is a straight line graph.

The equation 6.2 shows that m is proportional to v .

When $v = 2f$, $m = -1$.

Apparatus:

A convex lens
 A piece of card with narrow triangle shaped slit
 A screen
 A light source
 A metre rule
 A lens holder
 Plasticine

Procedure:**Figure 6.1**

1. Use the convex lens to focus a distant object such as a tree outside the laboratory on a screen. The distance between the screen and the lens is the estimated focal length, f_o of the lens. Record the estimated focal length f_o .

Note: *Make sure you get the best sharp image.*

2. Set up the apparatus as in **Figure 6.1**.
3. Measure and record the height, h_o of the triangular slit on the cardboard. This is the height of the object for this experiment.
4. Place the object in front of the lens at a suitable distance ($f_o < u < 2f_o$) and adjust the position of the screen so that a sharp image is projected on the screen.

5. Measure and record the object distance u , the image distance v and the height of the sharp image h_i .
6. Calculate the magnification of the image using

$$m = \frac{\text{height of image}}{\text{height of object}} = \frac{h_i}{h_o}.$$

Note: *Since the image is real and inverted, the value magnification m is negative.*

7. Change the location of the object. Repeat steps (5) and (6) for six sets of u , v and m .
8. Plot a graph of m against v .
9. Determine the focal length of the lens, f_l from the gradient of the graph.
10. Compare the values of f_o and f_l with the standard value. Write a comment.

REFERENCES

- Cutnell, J. D. Johnson, K. W. (2019). *Physics* (11th ed.). USA: John Wiley & Sons, Inc.
- Serway, R. A. & Jewett, J. A. (2019). *Physics for Scientists and Engineers* (10th ed.). International Student Edition. USA: Brooks/Cole Cengage Learning.
- Giancoli, D. C. (2016). *Physics – Principles with Application* (7th ed.). Prentice Hall.

ACKNOWLEDGEMENTS

The Matriculation Division, Ministry of Education Malaysia wish to thank everyone who has contributed in shaping and writing this PHYSICS LABORATORY MANUAL (5th Edition) for the Two Year Matriculation Programme. Special thanks go to those for their many valuable suggestions and conscientiousness in completing this manual.

Dr. Hajah Rosnarizah binti Abdul Halim
Director of Matriculation Division

Haji Mohd Yusof bin Samad
Deputy Director of Matriculation Division (Academic Sector)

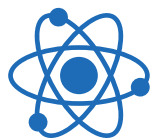
Mohd Junaidi bin Abd Aziz
Senior Principal Assistant Director
Matriculation Division

Jamilah binti Abdul Rashid
Aifaa binti Awang Kechik
Assistant Director

Reviewers for the Fifth Edition

- Haji Marhalim bin Hashim, KMPP
- Hayati binti Ibrahim, KMPh
- Tengku Ainun Marshida binti Tengku Hassan, KMM
- Nurul Zawani binti Zainudin, KMkt
- Dr. Syarifah Norsuhaila binti Syed Mahmud, KMPh
- Mohd Rohit bin Safuan, KMS
- Mohd Fauzi bin Abdul Dzubir, KMPk
- Rusman Rizal bin Abdul Rahman, KMPk
- Shamsuhanizul Bin Shamsudin, KMNS
- Hussaini bin Abd Rahman, KMK
- Che Mohamad Zamri bin Bakar, KMPk
- Hasimah binti Husin, KMK
- Nazaruddin bin Zakaria, KML
- Shazrina binti Abd Rahim, KMJ

Cover designed by Syed Nassir bin Syed Ahmad, KML



DP014 & DP024

e ISBN 978-983-2604-67-9



9 789832 604679